

Optimization of mechanical and water barrier properties of avocado seed starch based film and its application as smart pH indicator by adding blue butterfly pea flower extract

Ayşe Gul Asikkutlu, Meral Yildirim-Yalcin^{*}

Istanbul Aydın University, Engineering Faculty, Food Engineering Department, 34295 Istanbul, Türkiye

ARTICLE INFO

Keywords:

Avocado seed starch
Food waste
Optimization
Intelligent packaging

ABSTRACT

This study investigated the properties of films based on avocado (*Persea americana*) seed starch. A full factorial experimental design was performed using different amounts of starch (3–5 %) and glycerol (0.75–1.75 %) to investigate the mechanical, water solubility and swelling, water vapor permeability, and opacity properties of the films. The optimum film formulation with maximum mechanical strength, minimum water solubility and water vapor permeability was found to be 4 % starch and 1 % glycerol. An application study was conducted with blue butterfly pea flower (*Clitoria ternatea*) extract added at 10 % by volume of the film solution to test its use as a spoilage indicator in packaging chicken breast meat. The deterioration of chicken breast meat was monitored using pH changes and total mesophilic aerobic bacteria count. During storage at room temperature, the film's color transitioned from blue to green as the meat deteriorated, correlating with pH changes and microbial growth.

1. Introduction

Avocado fruit (*Persea americana* Mill.) grows naturally in Mexico and Central America. In recent years, avocado consumption has increased worldwide due to its high nutritional value and numerous health benefits. Avocado is rich in unsaturated fats, proteins, vitamins, minerals and many bioactive compounds, as well as having a low sugar content. It can be used in different sectors such as culinary, cosmetics and pharmacy (Wang et al., 2022). Global avocado consumption and trade is increasing and expected to increase in the near future (Hassan et al., 2024). As a result of increasing consumption and usage rates, its agricultural waste is also increasing. Avocado seeds, which are approximately 16 % of the dry weight of the fruit, are the most important waste (Hassan et al., 2024; Wang et al., 2022). Evaluating these seeds not only provides economic gain but also provides a solution to the environmental pollution that the seeds will create (Oluwajuyitan, 2024; Wang et al., 2022). Starch can be obtained from avocado seeds with a yield of over 60 % by dry weight (Tefaye et al., 2018).

Starch is an important biopolymer because it is cheap, abundant, biodegradable and non-toxic. Starch obtained from various sources such as wheat, corn, and cereals are used as a thickener and stabilizer in various industries such as food, cosmetics, pharmaceuticals, and textiles

(Araújo et al., 2020). Avocado seed is considered as a natural and economically promising source for starch with its characteristics (Builders et al., 2010; Chel-Guerrero et al., 2016). There are studies examining starch extraction from waste avocado seeds for different purposes such as textile applications, aerogel applications, biofoam production, encapsulation of bioactive compounds (Pires et al., 2024) and film production (Jimenez et al., 2021).

In recent years, bioplastic alternatives derived from biodegradable polymers have been produced to address the serious environmental problems caused by waste from petroleum-based plastic packaging (Jha & Kumar, 2019). It is estimated that 520,000–720,000 tons of avocado seeds are produced each year in the world (Tefaye et al., 2022). This amount will also increase with increasing consumption rates; therefore, researchers have been working on the use of starch obtained from avocado seeds as biopolymer packaging.

Plasticizers, such as glycerol or sorbitol, are typically incorporated into the film matrix to address the fragility and brittleness of pure starch films. The flexibility and processability of the films can be enhanced by the use of plasticizers, which can reduce internal starch hydrogen bonds (Muñoz-Gimena et al., 2024). Many researchers have used avocado seed starch (ASS) as a reinforcement in the biopolymer matrix to create films. For instance, Lubis et al. (2018) generated an ASS film that contained

^{*} Corresponding author.

E-mail address: meralyildirimyalcin@aydin.edu.tr (M. Yildirim-Yalcin).

microcrystalline cellulose from sugar palm kernel. Solihat et al. (2023) combined ASS with nanochitosan; and Muñoz-Gimena et al. (2024) employed nanocrystalline starch that was obtained through acid hydrolysis; Thanh et al. (2024) used ASS in combination with nanocellulose fibers. Moreover, Castro-García et al. (2022) used avocado oil and ethylene glycol as crosslinkers in ASS film. They determined that the thermal stability of the films was enhanced by crosslinking. Also, some researchers used ASS and Polylactic Acid (PLA) together to produce films (Dewi et al., 2024; Zúniga et al., 2024). Moreover, Waday and Aklilu (2023) developed an antimicrobial film by adding orange peel extract to a film prepared with ASS to increase the shelf life of food. However, none of these studies have investigated the amount of ASS and plasticizer used in film formulations. There is a lack of information on optimizing film formulations formed with ASS and plasticizers considering film properties.

Additional properties are gained by adding various active compounds to biodegradable films. It is very important to determine the potential of films produced with macromolecules extracted from food wastes as carriers for these active ingredients. Anthocyanins, which are natural pigments, are responsible for various colors such as red, orange, blue and purple and are used in the development of pH-sensitive color indicator materials. pH indicators are used in smart packaging to provide information about the quality, freshness and safety of the product in the package (Bhargava et al., 2020). The extract of butterfly pea flower (*Clitoria ternatea*) is abundant in anthocyanins and aromatic acyl compounds, which are responsible for its distinctive blue color and is particularly suitable for use in biodegradable pH-indicator smart packaging (Hashim et al., 2022). The primary anthocyanins found in blue butterfly pea flowers are polyacylated derivatives of delphinidin 3,3',5'-triglucoside, which are known as ternatins (Vidana Gamage et al., 2021).

Therefore, the objective of this study is to examine the impact of incorporating ASS and glycerol into the film formulation. To obtain this objective, full factorial design was employed to optimize the concentration of film components (ASS: 3–5 % and glycerol: 0.75–1.75 %) in order to produce a film formulation with the minimum water solubility and water vapor permeability and the maximum mechanical properties. Furthermore, the other objective is to investigate the feasibility of incorporating blue butterfly pea flower (BBF) extract into the optimum formulated film as an intelligent pH indicator in the storage of chicken breast meat.

2. Materials and methods

2.1. Materials

Avocado seeds of Hass and Bacon varieties were obtained from Mediterranean Natural Life Association (Antalya, Türkiye) and stored at -18°C until used. Sodium metabisulphite ($\text{Na}_2\text{S}_2\text{O}_5$) was purchased from Labor Teknik (Tekkim, Türkiye); Glycerol (84–88 %) was purchased from Sigma-Aldrich (Steinheim, Germany). Blue butterfly pea flower (*Clitoria ternatea*) was ordered online from a local market in dried form.

2.2. Starch extraction from avocado seeds

The method suggested by Silva et al. (2017) was applied with some modifications in starch extraction from avocado seeds. Frozen avocado seeds were thawed in an oven at 60°C for 20 min. The thin shell on the seeds was peeled off with a knife and cut into small pieces of approximately 0.5 cm and kept in sodium metabisulfite solutions at 5 % w/v concentration for 1 week. This concentration was obtained as a result of preliminary analyses. Then, it was ground in a laboratory blender and homogenized. The homogeneous mixtures were washed by passing through a $63\text{ }\mu\text{m}$ sieve with distilled water several times, and the obtained solutions were filtered through a filter paper with the help of a

vacuum until the filtrate became clear. The slurry was dried in a 40°C oven for 4 h and was passed through a $63\text{ }\mu\text{m}$ sieve. The obtained starch was stored at room temperature in a sealed plastic container.

2.2.1. Fourier transform infrared (FT-IR) spectroscopy

The FT-IR spectrum of the starch was measured using Bruker Invenio-S (Bruker Co., Germany) spectroscopy in the wavenumber range of $400\text{--}4000\text{ cm}^{-1}$ with a resolution of 4 cm^{-1} .

2.3. Experimental design

The experimental design was made with a three-level two factor (3^2) full factorial design. In the design, the independent variables were selected as the amount of ASS and glycerol. Three different levels were examined for each independent variable: 3, 4, and 5 g for ASS and 0.75, 1.25, and 1.75 g for glycerol. The experimental design was carried out by Design Expert Software (Version 11.1.0, State-Ease, Inc. Minneapolis, USA). The effects of the independent variables on the dependent variables (tensile strength, maximum elongation at break, water solubility, water swelling, water vapor permeability and opacity characteristics of films) were examined with this software and the response surface methodology analysis was performed and 3-dimensional surface plots were drawn. The data obtained for the dependent variables was analyzed using the second-order polynomial eq. (1) given below:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{11} x_1^2 + \beta_{22} x_2^2 + \beta_{12} x_1 x_2 \quad (1)$$

Y is the predicted response, x_1 and x_2 are the codes of the independent variables, β_0 is the constant, β_1 and β_2 are linear, β_{11} and β_{22} are quadratic and β_{12} is the interaction coefficients.

2.4. Preparation of edible film from ASS

Edible films were prepared by casting method using extracted ASS, glycerol as plasticizer and distilled water. The ASS (4, 5, and 6 g) was weighed (AND GR200, Japan) into a beaker and 100 mL of distilled water was added. The film solution in the beaker was mixed in a water bath at 90°C (Stuart SWB2D, UK) with a mechanical stirrer (IKA RW20, Germany) at 400 rpm for 30 min. After the solution cooled to room temperature, different amounts of glycerol (0.75, 1.25, and 1.75 g) were added. The mixture was homogenized with the mechanical stirrer at 400 rpm for 15 min. Then it was kept in an ultrasound water bath (PMUY4LD, Protech, Istanbul, Türkiye) for 10 min to remove air bubbles. It was poured (15 mL) into 12 cm inner diameter Teflon petri dishes and dried in a 50°C oven (ED53, Binder, Germany) for 24 h. After drying, the films were removed from the dishes and kept in a desiccator containing saturated $\text{Mg}(\text{NO}_3)_2$ (55 % relative humidity) at 25°C for 48 h (Yildirim-Yalcin et al., 2021a).

2.5. Film analyses

2.5.1. Mechanical properties

The maximum elongation at break (EAB) and the tensile strength (TS) were measured in accordance with the ASTM (2012) standard method under ambient conditions. A rectangular film sample ($15\text{ mm} \times 70\text{ mm}$ was cut from the films) was placed vertically between the two grips of the mechanical testing machine (LAB3-2512 A, Mesdan LAB, Italy) with 40 mm initial length. The tests were carried out with a load cell of 100 N and a crosshead speed of 25 mm/min (Kurtfaki & Yildirim-Yalcin, 2023).

2.5.2. Water solubility

The film samples were cut into circular shapes (20 mm diameter), placed in glass petri dishes and left to dry in a vacuum oven (EV018, Nuve, Ankara, Türkiye) at 80°C for 24 h. The dried films were then kept in 50 mL of distilled water at 25°C with continuous stirring (SSL1,

Table 1

Experimental design and experimental data.^a

Run	Variables					Responses		
	ASS (g)	Glycerol (g)	TS (N/mm ²)	EAB (%)	Solubility (%)	Swelling (g/g)	WVP (gmm/m ² hPa)	Opacity (Abs _{600nm} /mm)
1	4	0.75	1.48 ± 0.09	3.38 ± 0.18	13.32 ± 1.49	1.00 ± 0.03	0.84 ± 0.00	10.44 ± 0.77
2	4	1.25	0.37 ± 0.03	21.38 ± 2.83	19.52 ± 0.34	0.67 ± 0.05	0.69 ± 0.02	9.58 ± 1.98
3	4	1.75	0.21 ± 0.01	28.50 ± 1.06	26.44 ± 0.23	0.66 ± 0.09	0.96 ± 0.05	6.59 ± 0.17
4	5	0.75	1.74 ± 0.05	3.10 ± 0.28	17.69 ± 1.45	0.84 ± 0.03	1.24 ± 0.18	9.08 ± 0.21
5	5	1.25	0.53 ± 0.01	18.00 ± 1.06	17.17 ± 1.35	0.70 ± 0.08	1.06 ± 0.02	8.09 ± 0.51
6	5	1.75	0.30 ± 0.00	28.00 ± 1.77	37.86 ± 1.21	0.54 ± 0.00	1.43 ± 0.01	3.83 ± 0.12
7	6	0.75	2.23 ± 0.32	0.75 ± 0.01	12.13 ± 1.45	0.99 ± 0.15	1.14 ± 0.01	7.64 ± 0.58
8	6	1.25	1.24 ± 0.05	7.58 ± 0.11	13.59 ± 1.21	0.80 ± 0.01	1.26 ± 0.05	8.02 ± 0.63
9	6	1.75	0.46 ± 0.02	15.25 ± 1.41	33.47 ± 3.29	0.45 ± 0.08	1.42 ± 0.13	4.28 ± 0.03

TS: Tensile strength, EAB: Elongation at break, WVP: Water vapor permeability.

^a Values were given as means ± standard deviation.

Stuart, UK) for 24 h. The insoluble content was separated and dried in a vacuum oven at 80 °C for 24 h before weighting (final weight). Finally, the percent solubility of the films in water was determined by dividing their soluble quantities by the initial dry weights.

2.5.3. Swelling

The dry weight of the circular shaped subsamples of films were initially weighed and submerged in water at 25 °C for 30 min with continuous stirring (SSL1, Stuart, UK). Then, the samples were dried on filter paper to eliminate excess water and weighed. The ratio of g water gained to initial weight of the film was used to calculate the swelling (Kurtfaki & Yildirim-Yalcin, 2023).

2.5.4. Water vapor permeability

The modified gravimetric water method was employed to determine the water vapor permeability coefficient (WVP) of films, which was based on the standard method E96/E96M (ASTM, 2015). The experiment was conducted in accordance with the stages of the procedure that were previously described (Yildirim-Yalcin et al., 2019). The results were determined in g mm/m² h kPa.

2.5.5. Opacity

In order to determine the opacity value of the film, absorbance measurement was made at 600 nm using a UV-Visible spectrophotometer (T60UV, PG instruments, United Kingdom). The films to be measured were cut into rectangular pieces of 0.4 cm width and 3 cm height and placed on the outer surface of the spectrophotometer cuvettes, and the opacity was determined as the ratio of absorbance to film thickness (mm).

2.6. Statistical analysis and optimization and verification

Statistical parameters were determined by performing Variance Analysis (ANOVA) with the Design Expert program. Three-dimensional (3D) surface graphs were drawn by performing multiple regression analysis. The suitability of the models was tested with the determination of coefficient (R²) and lack of fit values. The optimum film formulation was calculated numerically using the desirability function. Optimization conditions were determined as maximum tensile strength and maximum elongation at break, minimum water solubility and water vapor permeability, and water swelling and opacity values were kept within the obtained value range.

The film in the formulation obtained as a result of the optimization study (4 g ASS and 1 g glycerol) was prepared for verification purposes. The estimated data determined at the optimum point and the experimental data obtained after the verification study were compared with the *t*-test using SPSS (IBM SPSS Statistics 19, SPSS Inc., Chicago, USA) software (*p* = 0.05). Experiments were performed at least in duplicate and the experimental data was expressed as mean ± standard deviation.

2.7. Addition of BBF extract to the optimum film

2.7.1. Extraction of BBF

BBF samples purchased dried form were ground with a laboratory blender and passed through a 100 µm sieve. Obtained powder (10 g) was mixed with 100 mL of distilled water (1:10 w/v) and kept in an ultrasound water bath at room temperature for 1 h. The suspension was vacuum filtrated, and the extract was stored at −80 °C away from light and air.

2.7.2. Preparation of ASS film enriched with BBF extract

Film containing 4 g ASS and 1 g glycerol at the optimum point was prepared as described in Section 2.4. Differently, starch was gelatinized with 90 mL distilled water, and this solution was homogenized by adding glycerol and BBF extract (10 mL) after cooling to room temperature. The remaining procedure was followed as described.

In addition, in order to examine the molecular structures of all films, FT-IR spectra were measured as described in Section 2.2.1.

2.7.3. pH Sensitivity of the Film

To examine the color response of the films containing BBF extract at different pHs, approximately 2 cm × 2 cm films were cut and soaked in buffer solutions at different pHs. Images of the films were taken with a camera after 10 min.

2.7.4. Packaging test

The films containing BBF extract were used to monitor the deterioration of chicken breast meat during storage. For this purpose, 10 g of chicken breast meat was placed in a sterile petri dish. The square pieces of the films (approximately 2.5 cm × 2.5 cm) were attached to the petri lid with small tapes without any contact with chicken meat. Chicken meat samples were stored at 25 °C for 3 d under room conditions. The color changes in the films were monitored by taking photographs every day. Additionally, the pH value and the Total Mesophilic Aerobic Bacteria (TMAB) count of chicken meat samples were measured during the 3-d storage. The chicken meat samples (10 g) and 90 mL of sterile Ringer's solution were homogenized under aseptic conditions with a stomacher. Dilutions were prepared and 0.1 mL of dilutions were transferred and spread on Plate Count Agar (PCA). After incubation (24 h at 37 °C), the TMAB counts were given as log₁₀ colony forming unit (CFU)/g. Also, the pH measurement of the homogenates was measured using a pH meter.

2.7.5. Statistical analysis

In order to determine the differences between the properties of optimum film and optimum film containing BBF extract, ANOVA was performed using SPSS (IBM SPSS Statistics 19, SPSS Inc., Chicago, USA) software. In addition, the differences between different storage days during the spoilage test of chicken meat were determined by Duncan's multiple comparison test (*p* = 0.05). Experiments were performed at least in duplicate, and the experimental data was expressed as mean ±

Table 2

Analysis of variance and regression coefficients of the second order polynomials for all responses.

	TS (N/mm ²)	EAB (%)	Solubility (%)	Swelling (g/g)	WVP (gmm/m ² h kPa)	Opacity
Model	Quadratic	Quadratic	Quadratic	Quadratic	Quadratic	Quadratic
p-value	< 0.0001	< 0.0001	< 0.0001	0.0002	< 0.0001	< 0.0001
Lack of fit (p-value)	0.07	0.09	0.06	0.07	0.13	0.11
Determination of coefficient (R ²)	0.97	0.96	0.91	0.84	0.91	0.92
Adjusted-R ²	0.96	0.94	0.87	0.77	0.87	0.88
Predicted-R ²	0.93	0.91	0.78	0.62	0.80	0.82
Regression coefficients ^a						
Intercept	β ₀	18.03	19.59	0.68	1.13	7.96
Linear	β ₁	−4.95***	−0.27	−0.014	0.22***	−1.04***
	β ₂	−0.75***	10.75***	−0.20***	0.098**	−2.08***
Quadratic	β ₁₁	0.14	−3.56*	−0.05	−0.19**	0.69
	β ₂₂	0.36***	−2.49	6.98**	0.070	−1.44**
Interaction	β ₁₂	−0.12*	−2.65*	−0.021	0.042	0.12

TS: tensile strength, EAB: elongation at break, WVP: water vapor permeability.

Significance level: *p < 0.05, **p < 0.01, ***p < 0.001.

^a Coefficients of Eq. (1). β₀: intercept; β₁ and β₂ linear regression coefficients for avocado seed starch (ASS) and glycerol; β₁₁ and β₂₂: quadratic regression coefficients for ASS x ASS and Glycerol x Glycerol; β₁₂ regression coefficients for interaction between ASS x Glycerol.

standard deviation.

3. Results and discussion

3.1. Model development

The ASS and glycerol concentrations were optimized in terms of mechanical, water solubility, swelling, water vapor permeability and opacity properties of films. The experimental design points and the data are given in Table 1. Table 2 demonstrates that the developed models were statistically significant for all responses ($p < 0.001$). High R² (0.97–0.84), adjusted-R² (0.96–0.77), and non-significant lack of fit values indicated the adequacy of models.

3.2. Effect of ASS and Glycerol Concentration on Responses

3.2.1. Mechanical properties

When Table 1 is examined, the TS parameters of the films were between 0.21 ± 0.01 and 2.23 ± 0.32 N/mm² and the EAB parameters were between 0.75 ± 0.01 % and 28.50 ± 1.06 %. According to the ANOVA table (Table 2), the quadratic models developed for TS and EAB are significant ($p < 0.0001$) with insignificant model incompatibility. The effect of ASS and glycerol concentration on TS and the quadratic effect of glycerol are highly significant, while their interactions are less significant. ASS and glycerol concentrations have highly significant effects on EAB. The quadratic effect of ASS and the interaction between ASS and glycerol result in less significant effects.

Three-dimensional graphs were drawn according to the regression equations (Fig. 1). When Table 1, Table 2, and Fig. 1 are examined, it is seen that the TS values of the films are positively correlated with the ASS concentration and negatively correlated with the glycerol concentration ($p < 0.001$). In other words, as the amount of glycerol increases, the TS of the films decreases. When the EAB data are examined, a significant decrease is observed as the amount of starch increases ($p < 0.001$). In addition, when the amount of glycerol increases, the EAB increases significantly ($p < 0.001$). Plasticizers are components that interact with starch molecules and make the material flexible. With plasticizers, TS decreases but elongation increases (Jansson & Järnström, 2005). Glycerol reduces the TS of the film due to its small size, hydroxyl groups and hydrophilic structure. It attracts water with its hydrophilic nature; water enters between the molecules and increases the distance between them, thus decreasing the TS (Santacruz et al., 2015).

3.2.2. Solubility and swelling

The optimization study aimed to minimize water solubility, which is an important characteristic that provides information about the water

sensitivity of films. According to Table 1, the solubility of the films varied between 12.13 ± 1.45 % and 37.86 ± 1.21 %. The 3D surface plots in Fig. 1 show how two independent variables affect the solubility property. While increasing the amount of ASS increased the solubility up to a certain level, it caused a decrease after the peak point. However, ASS concentration had no significant effect on the solubility of the films. Increasing glycerol concentrations increased the solubility of the films for all starch concentrations ($p < 0.001$). Because plasticizers positively affect the water solubility of films with their hydrophilic properties (Cuq et al., 1997). In addition, the quadratic effects of ASS and glycerol were significant.

Another feature that demonstrates the films' resistance to water is their swelling when submerged. The swelling is a feature that should be considered during the packaging of high-moisture foods. Furthermore, improved swelling properties of films are important in controlled release studies. Therefore, the study was conducted to ensure the swelling properties of films stayed within the experimental range while determining the optimization parameters. The swelling values of the films in water ranged between 0.45 ± 0.08 and 1.00 ± 0.03 g water/g film (Table 1). Table 2 and Fig. 1 demonstrate that the ASS concentration did not significantly affect the swelling. Glycerol concentration significantly affected it. The swelling values of the films in water decrease when the amount of glycerol increases. The inclusion of plasticizer molecules in the film matrix ensures that there is not enough space left to absorb water molecules when it encountered water, and thus it is thought that increasing the amount of glycerol can reduce swelling (Tabatabaei et al., 2022).

3.2.3. Water vapor permeability

The optimization parameters for ASS films aimed to identify a film formulation with a low water vapor pressure (WVP), as this would accurately reflect the water vapor barrier properties of the film. The WVP of the film samples ranged from 0.69 ± 0.02 to 1.43 ± 0.01 gmm/m²hkPa (Table 1). There was a correlation between film WVP and starch and glycerol concentrations, as well as quadratic effects. In other words, increasing the amount of starch and glycerol caused the WVP of the films to increase. Similarly, Mali et al. (2004) reported an increase in the WVP with the increased amount of yam starch. This result was explained as the excess amount of starch increased the number of free hydroxyls in the film matrix, increasing the interaction with water and finally the WVP. Also, glycerol can increase the WVP of the films by increasing the distance between the starch chains (Yildirim-Yalcin et al., 2021b).

3.2.4. Opacity

The opacity of the films varied between 3.83 ± 0.12 and 10.44 ± 0.77 . The opacity values of the films did not change in a wide range and

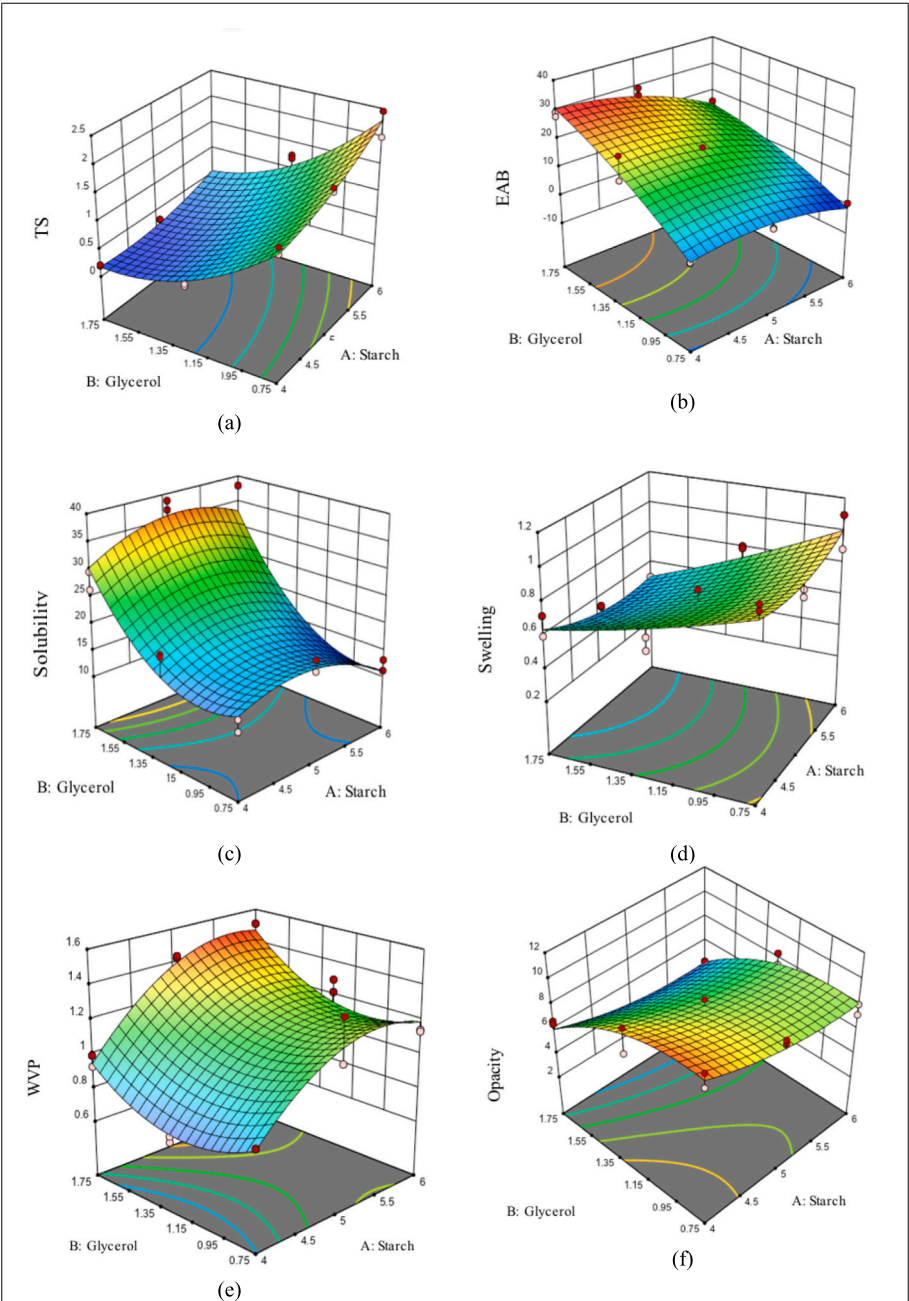


Fig. 1. Response surface diagrams showing the effects of ASS and glycerol concentration on tensile strength (N mm^{-2} ; a), elongation at break (%; b), solubility (c), swelling (d), water vapor permeability (c) and opacity (f) of edible films. ASS: avocado seed starch.

not critical in terms of strength properties. Therefore, when determining the optimum point, the opacity value was selected to be within the range of the results obtained. When the values in Table 2 and Fig. 1 are examined, increasing the amount of starch and glycerol caused a significant decrease in the opacity of the films ($p < 0.05$).

3.3. Optimization and validation

Simultaneous optimization analysis for multiple responses was performed numerically using the Desirability function. Optimization conditions were set to ensure TS and EAB were maximum, water solubility and WVP were minimum. The optimum point was determined with a desirability value of 0.56, and a content of 4 g ASS and 1 g glycerol. A verification study was carried out at the optimum point, and all analyses were also performed. Estimated model data and experimental data are

Table 3
Predicted and experimental values at the optimum conditions for avocado seed starch-based film.

Responses	Predicted value (mean)	Experimental value (mean $\pm$ standard deviation)
TS (N/mm^2)	0.85	0.56 ± 0.04
EAB (%)	12.18	15.00 ± 2.12
Solubility (%)	13.52	14.32 ± 3.18
Swelling (g/g)	0.85	0.84 ± 0.02
WVP ($\text{gmm/m}^2\text{h kPa}$)	0.73	0.71 ± 0.01
Opacity	10.42	8.56 ± 0.31

TS: tensile strength, EAB: elongation at break, WVP: water vapor permeability.

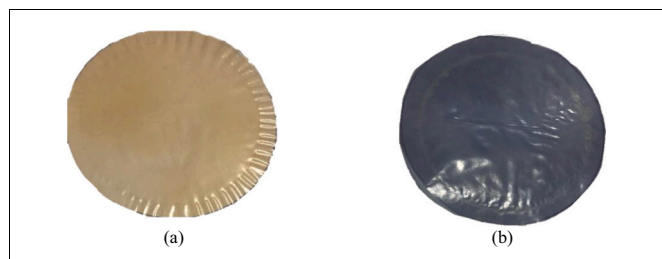


Fig. 2. Images of (a) optimum formulated film and (b) optimum formulated film with BBF extract.

Table 4

Properties of optimum film containing BBF extract.

Properties	Results
TS (N/mm ²)	0.39 ± 0.00
EAB (%)	29.63 ± 0.53
Solubility (%)	19.70 ± 0.19
Swelling (g/g)	1.05 ± 0.10
WVP (gmm/m ² hPa)	0.71 ± 0.09
Opacity	9.30 ± 0.56

TS: tensile strength, EAB: elongation at break, WVP: water vapor permeability.

given in Table 3. The *t*-test results showed that there were no significant differences between the estimated values and the experimental values ($p > 0.05$). In this case, it can be said that the experimental study confirmed the estimated model parameters.

3.4. The properties of film enriched with BBF extract

In order to evaluate the smart packaging production potential of the film in the optimum formulation, BBF extract was added to the film. The images of the films produced at the optimum point are given in Fig. 2. The findings of the BBF extract containing film properties are given in

Table 4. The results of the optimum film and the optimum film containing the BBF extract were compared statistically. It was found that the addition of the extract caused a significant decrease in the TS and a significant increase in the EAB, water solubility and swelling values ($p < 0.05$). No significant effect was observed in WVP and opacity results. Generally, substances other than crosslinking agents that are added to polymer films reduce the TS of the films by reducing the molecular interactions between the polymer chains (Kanmani & Rhim, 2014). In addition, extracts can act as plasticizers in the film matrix, increasing chain mobility and elongation (Bodini et al., 2020). Researchers found that adding grape seed extract to arrowroot starch film increased the film's water solubility. The water-soluble extract increased the soluble component of the film, and the extracts acted as plasticizers in the film (Nogueira et al., 2022).

3.5. FT-IR spectroscopy

The functional groups of ASS and films were identified using FT-IR spectroscopy (Fig. 3). The ASS spectrum was found to be similar to the typical starch spectrum. There are two characteristic regions in the starch spectrum. The first is the -OH stretching vibration, in the range of 3650–3000 cm⁻¹. The second is the major adsorption bands region, known as the fingerprint region, located in the range of 1200–800 cm⁻¹ (Pozo et al., 2018). Similar peaks were observed in ASS and ASS films. The -OH stretching vibration of the glucose unit is in the range of 3650–3000 cm⁻¹. Moreover, the peak observed around 2900 cm⁻¹ corresponds to the C–H stretching of methyl groups. The peak around 1645 cm⁻¹ represents the H₂O bending vibration and the other peak around 1728 cm⁻¹ represents C=O bending vibration. Peaks in the 1150–1000 cm⁻¹ region are related to C–O and C–C stretching with C–O–H contributions (Araújo et al., 2020; Castro-García et al., 2022; Wijaya et al., 2019).

3.6. Chicken packaging test

The color changes of BBF extract and the films produced using this

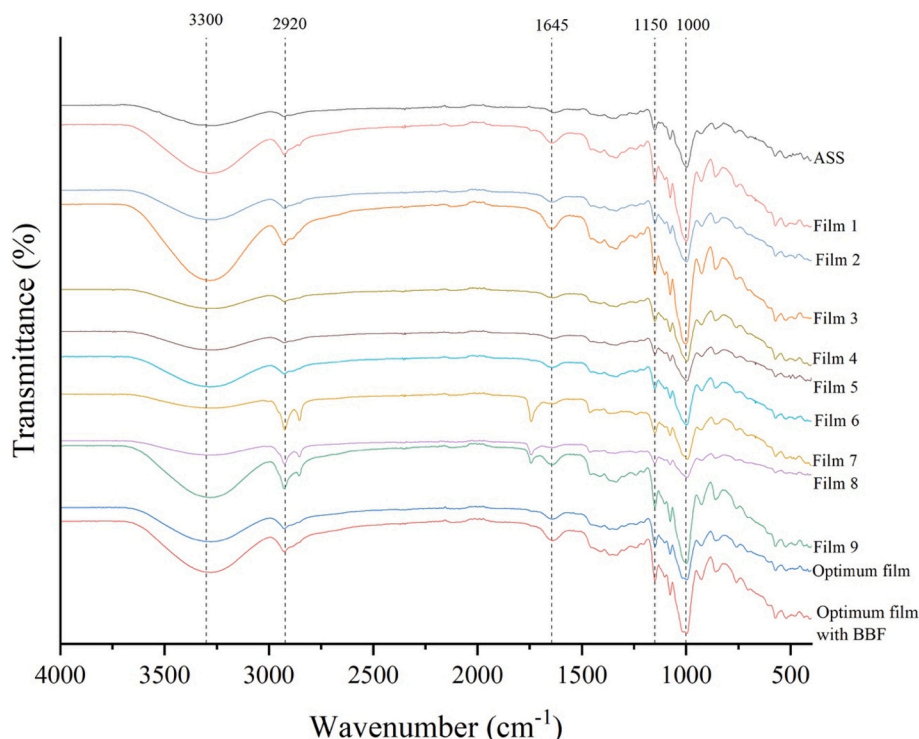


Fig. 3. FT-IR spectra of films.

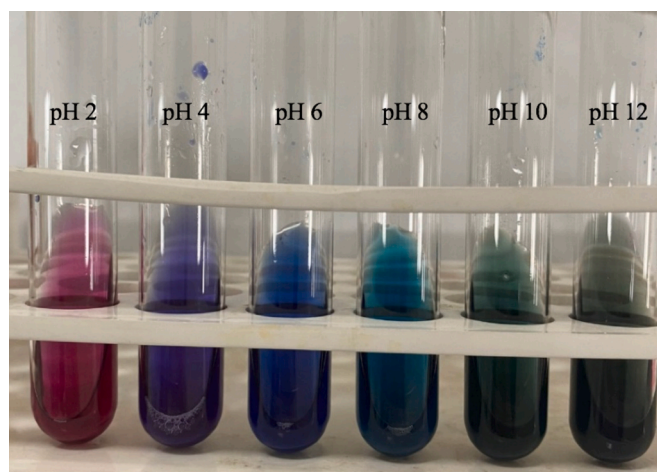


Fig. 4. Color change of BBF extract at different pH.

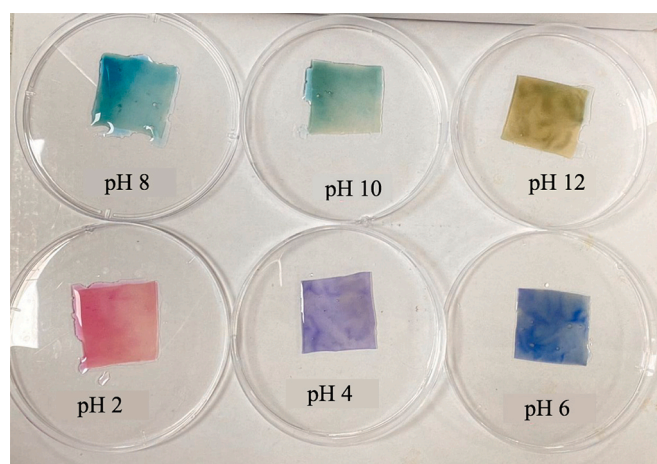


Fig. 5. Color change of ASS film with BBF extract at different pH.

Table 5
pH and TMAB count during the spoilage of chicken breast meat.

Storage Day	pH	TMAB count ($\log_{10}$ CFU/g)
0	5.65 ± 0.01^d	3.27 ± 0.19^e
1	6.11 ± 0.01^c	7.07 ± 0.03^b
2	6.96 ± 0.01^b	9.36 ± 0.04^a
3	7.19 ± 0.01^a	9.75 ± 0.28^a

extract at different pHs are shown in Fig. 4 and Fig. 5, respectively. There were significant color changes in the extracts at various pH values: fuchsia pink at pH 2, purple at pH 4, saxe blue at pH 6, petrol blue at pH 8, petrol green at pH 10, and jungle green at pH 12. It was observed that the optimum films with BBF extract exhibit a significant color change depending on the pH change. This demonstrates that the films can be used as a pH-sensitive color indicator.

The use of BBF extract-containing films as pH indicator smart packaging during the spoilage of chicken breast meat was studied. As the storage period lengthens, the microorganisms typically grow rapidly, producing various metabolites that lead to unpleasant odor, changes in taste, color and texture, and ultimately, sensory rejection. The breakdown of amino acids, which leads to the formation of volatile nitrogenous compounds like ammonia and amines, primarily causes degradation in meat and seafood and increases the pH inside the package (Hernández-Jover et al., 1996). TMAB count and pH changes during storage in freshly packaged chicken breast samples are given in Table 5.

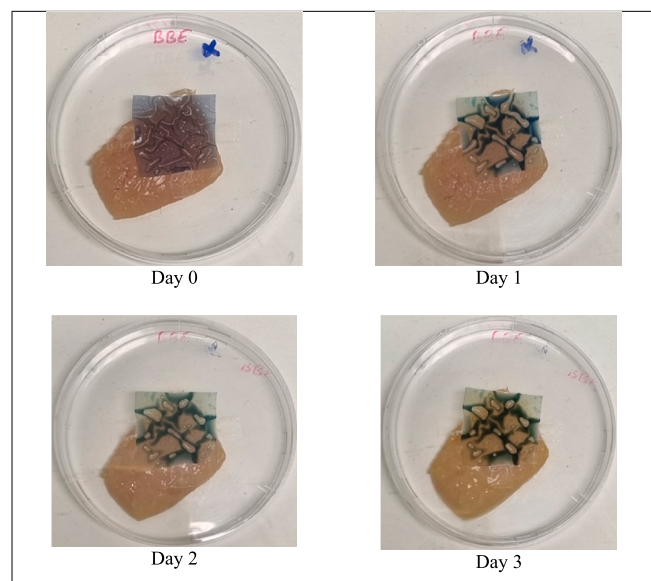


Fig. 6. Packaging tests of ASS films with BBF extract of chicken breast meat spoilage for 3 d of storage at 25 °C.

TMAB counts increased as the storage period increased. The accepted limit for TMAB count in poultry products was reported by Senter et al. (2000) as 7–8 log CFU/g. The TMAB count of chicken meat samples reached its critical value on the first d. Also, it was seen that pH increased as the storage period increased. As illustrated in Fig. 6, depending on the pH change, there was a change in the films from blue to petrol blue and finally turning green on the final day. Therefore, the color change of the ASS film containing BBF extract from blue to green in the package indicates the deterioration of the chicken meat.

4. Conclusion

The effects of ASS and glycerol concentration on the mechanical and water barrier properties of the films were determined by RSM. Three-dimensional surface graphs and mathematical models showed that the concentration of film components affected the film properties ($p < 0.05$). The amount of ASS in the films showed a positive relationship with both TS and WVP. The glycerol concentration showed negative effects on TS and swelling in water, while it had positive effects on water solubility and WVP. The optimization study determined that the film containing 4 g ASS and 1 g glycerol in 100 mL solution had the highest TS and EAB, and the lowest WVP and water solubility properties.

Moreover, BBF extract was added to the optimum film formulation, and its use as a pH indicator smart packaging in the chicken breast meat storage was investigated. When the properties of the new film were compared with the optimum film, it was found that the addition of the extract caused a significant decrease in the TS and a significant increase in the maximum EAB, water solubility, and swelling values. The WVP and opacity results showed no significant effect. The films were attached to the packages containing chicken breast meats, and the TMAB count and pH changes were determined during 3 d of storage at room temperature, and the color changes in the films were observed. The results showed that a visible color change was seen in the films as the chicken breast fillet deteriorated. Therefore, this kind of film can be recommended as smart packaging to monitor the changes due to deterioration in meats during storage and distribution.

CRedit authorship contribution statement

Ayşe Gul Asikkutlu: Writing – original draft, Methodology, Formal analysis, Data curation. Meral Yildirim-Yalcin: Writing – review &

editing, Writing – original draft, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was derived from the master thesis of Ayse Gul Asikkutlu.

References

- Araújo, R. G., Rodríguez-Jasso, R. M., Ruiz, H. A., Govea-Salas, M., Rosas-Flores, W., Aguilar-González, M. A., ... Aguilar, C. N. (2020). Hydrothermal-microwave processing for starch extraction from Mexican avocado seeds: operational conditions and characterization. *Processes*, 8(7), 759.
- ASTM. (2012). *Standard test method for tensile properties of thin plastic sheeting - D882-12*. ASTM. West Conshohocken, PA: Annual Book of American Standard Testing Methods.
- ASTM. (2015). *Standard test method for water vapor transmission of materials - E96/E96M-15*. West Conshohocken, PA: ASTM, Annual Book of American Standard Testing Methods.
- Bhargava, N., Sharanagat, V. S., Mor, R. S., & Kumar, K. (2020). Active and intelligent biodegradable packaging films using food and food waste-derived bioactive compounds: A review. *Trends in Food Science & Technology*, 105, 385–401.
- Bodini, R. B., Pugine, S. M. P., de Melo, M. P., & de Carvalho, R. A. (2020). Antioxidant and anti-inflammatory properties of orally disintegrating films based on starch and hydroxypropyl methylcellulose incorporated with *Cordia verbenacea* (erva baleeira) extract. *International Journal of Biological Macromolecules*, 159, 714–724.
- Builders, P. F., Nnurnum, A., Mbah, C. C., Attama, A. A., & Manek, R. (2010). The physicochemical and binder properties of starch from *Persea americana* Miller (Lauraceae). *Starch-Stärke*, 62(6), 309–320.
- Castro-García, P. G., Vasquez-García, S. R., Flores-Ramírez, N., Rico, J. L., Abdel-Gawwad, H. A., García-González, L., & Fernández-Quiroz, D. (2022). Polymeric films prepared from starch and a crosslinker extracted from avocado seeds. *Journal of Applied Polymer Science*, 139(31), Article e52725.
- Chel-Guerrero, L., Barbosa-Martín, E., Martínez-Antonio, A., González-Mondragón, E., & Betancur-Ancona, D. (2016). Some physicochemical and rheological properties of starch isolated from avocado seeds. *International Journal of Biological Macromolecules*, 86, 302–308.
- Cuq, B., Gontard, N., Cuq, J. L., & Guilbert, S. (1997). Selected functional properties of fish myofibrillar protein-based films as affected by hydrophilic plasticizers. *Journal of Agricultural and Food Chemistry*, 45(3), 622–626.
- Dewi, R., Sylvia, N., Zulnazri, Z., Fithra, H., Riza, M., Siregar, J. P., ... Anis, S. (2024). The Optimization of Avocado-Seed-Starch-Based Degradable Plastic Synthesis with a Polylactic Acid (PLA) Blend Using Response Surface Methodology (RSM). *Polymers*, 16(16), 2384.
- Hashim, S. B., Tahir, H. E., Liu, L., Zhang, J., Zhai, X., Mahdi, A. A., ... Jiyong, S. (2022). Intelligent colorimetric pH sensing packaging films based on sugarcane wax/agar integrated with butterfly pea flower extract for optical tracking of shrimp freshness. *Food Chemistry*, 373(131), 514.
- Hassan, S., Kabir, M. H., Mazhar, D., Imran, Z., Rehman, A., Afzal, S., & Bibi, S. (2024). Chemical processing and various applications of starch extracted from avocado seeds: a step towards sustainable food waste management. *Starch-Stärke*, 76(5–6), 2,300,114.
- Hernández-Jover, T., Izquierdo-Pulido, M., Veciana-Nogués, M. T., & Vidal-Carou, M. C. (1996). Biogenic amine sources in cooked cured shoulder pork. *Journal of Agricultural and Food Chemistry*, 44(10), 3097–3101.
- Jansson, A., & Järnström, L. (2005). Barrier and mechanical properties of modified starches. *Cellulose*, 12, 423–433.
- Jha, A., & Kumar, A. (2019). Biobased technologies for the efficient extraction of biopolymers from waste biomass. *Bioprocess and Biosystems Engineering*, 42, 1893–1901.
- Jimenez, R., Sandoval-Flores, G., Alvarado-Reyna, S., Aleman-Castillo, S. E., Santiago-Adame, R., & Velazquez, G. (2021). Extraction of starch from Hass avocado seeds for the preparation of biofilms. *Food Science and Technology*, 42, Article e56820.
- Kanmani, P., & Rhim, J. W. (2014). Development and characterization of carrageenan/grapefruit seed extract composite films for active packaging. *International Journal of Biological Macromolecules*, 68, 258–266.
- Kurtfaki, M., & Yildirim-Yalcin, M. (2023). Characterization of *Laurus nobilis* L. leaf essential oil incorporated maize starch and rice protein films. *Journal of Food Measurement and Characterization*, 17(5), 4954–4962.
- Lubis, M., Harahap, M. B., Ginting, M. H. S., Sartika, M., & Azmi, H. (2018). Production of bioplastic from avocado seed starch reinforced with microcrystalline cellulose from sugar palm fibers. *Journal of Engineering Science and Technology*, 13(2), 381–393.
- Mali, S., Grossmann, M. V. E., García, M. A., Martino, M. N., & Zaritzky, N. E. (2004). Barrier, mechanical and optical properties of plasticized yam starch films. *Carbohydrate Polymers*, 56(2), 129–135.
- Muñoz-Gimena, P. F., Aragón-Gutiérrez, A., Blázquez-Blázquez, E., Arrieta, M. P., Rodríguez, G., Peponi, L., & López, D. (2024). Avocado Seed Starch-Based Films Reinforced with Starch Nanocrystals. *Polymers*, 16(20), 2868.
- Nogueira, G. F., Soares, I. H. B. T., Soares, C. T., Fakhouri, F. M., & de Oliveira, R. A. (2022). Development and characterization of arrowroot starch films incorporated with grape pomace extract. *Polysaccharides*, 3(1), 250–263.
- Oluwajuyitan, T. D. (2024). Avocado seed starch: structure, functionality, and applications. *Non-Conventional Starch Sources*, 3–19.
- Pires, J. B., Dos Santos, F. N., da Cruz, E. P., Fonseca, L. M., Siebeneichler, T. J., Lemos, G. S., ... Dias, A. R. G. (2024). Starch extraction from avocado by-product and its use for encapsulation of ginger essential oil by electrospinning. *International Journal of Biological Macromolecules*, 254(127), 617.
- Pozo, C., Rodríguez-Llamazares, S., Bouza, R., Barral, L., Castaño, J., Müller, N., & Restrepo, I. (2018). Study of the structural order of native starch granules using combined FTIR and XRD analysis. *Journal of Polymer Research*, 25, 1–8.
- Santacruz, S., Rivadeneira, C., & Castro, M. (2015). Edible films based on starch and chitosan. Effect of starch source and concentration, plasticizer, surfactant's hydrophobic tail and mechanical treatment. *Food Hydrocolloids*, 49, 89–94.
- Senter, S. D., Arnold, J. W., & Chew, V. (2000). APC values and volatile compounds formed in commercially processed, raw chicken parts during storage at 4 and 13C and under simulated temperature abuse conditions. *Journal of the Science of Food and Agriculture*, 80(10), 1559–1564.
- Silva, I. R. A., Magnani, M., de Albuquerque, F. S. M., Batista, K. S., Aquino, J. D. S., & Queiroga-Neto, V. (2017). Characterization of the chemical and structural properties of native and acetylated starches from avocado (*Persea americana* Mill.) seeds. *International Journal of Food Properties*, 20(sup1), S2–S279.
- Solihat, I., Setyawati, S. R., Sulistiawaty, L., & Khoirurrizal, M. F. (2023). Synthesis and characterization of nano chitosan-avocado seed starch as edible films. *Jurnal Kimia Riset*, 8(1).
- Tabatabaei, S. D., Ghiasi, F., Gahruie, H. H., & Hosseini, S. M. H. (2022). Effect of emulsified oil droplets and glycerol content on the physicochemical properties of Persian gum-based edible films. *Polymer Testing*, 106(107), 427.
- Tesfaye, T., Ayele, M., Gibril, M., Ferede, E., Limeneh, D. Y., & Kong, F. (2022). Beneficiation of avocado processing industry by-product: A review on future prospect. *Current Research in Green and Sustainable Chemistry*, 5(100), 253.
- Tesfaye, T., Gibril, M., Sithole, B., Ramjugernath, D., Chavan, R., Chunilall, V., & Gounden, N. (2018). Valorisation of avocado seeds: extraction and characterization of starch for textile applications. *Clean Technologies and Environmental Policy*, 20, 2135–2154.
- Thanh, N. C., Duy, N. B. A., Xuyen, H. T. H., Huy, N. T., Dong, B. P., & Giang, T. Q. (2024). *Environmentally friendly bioplastic films based on avocado seeds starch and nanocellulose fibers* (pp. 1–11). Polymer Science, Series A.
- Vidana Gamage, G. C., Lim, Y. Y., & Choo, W. S. (2021). Anthocyanins from *Clitoria ternatea* flower: Biosynthesis, extraction, stability, antioxidant activity, and applications. *Frontiers in Plant Science*, 12(792), 303.
- Waday, Y. A., & Aklilu, E. G. (2023). ANN and RSM modeling for the synthesis of avocado seed starch combined orange peel extract antimicrobial packaging film. *International Journal of Polymeric Science*, 2023(1), 8,877,598.
- Wang, J., Li, Y., Jin, Z., & Cheng, Y. (2022). Physicochemical, morphological, and functional properties of starches isolated from avocado seeds, a potential source for resistant starch. *Biomolecules*, 12(8), 1121.
- Wijaya, C., Do, Q. D., Ju, Y. H., Santoso, S. P., Putro, J. N., Laysandra, L., ... Ismadji, S. (2019). Isolation and characterization of starch from *Limnophila aromatica*. *Heliyon*, 5 (5).
- Yıldırım-Yalcin, M., Sadıkoğlu, H., & Şeker, M. (2021a). Characterization of edible film based on grape juice and cross-linked maize starch and its effects on the storage quality of chicken breast fillets. *Lwt*, 142(111), 012.
- Yıldırım-Yalcin, M., Sadıkoğlu, H., & Şeker, M. (2021b). Optimization of mechanical and antioxidant properties of edible film based on grape juice and cross-linked maize starch and evaluation of its effects on the storage quality of fresh-cut pineapple. *Journal of Food Measurement and Characterization*, 15, 4669–4678.
- Yıldırım-Yalcin, M., Seker, M., & Sadıkoğlu, H. (2019). Development and characterization of edible films based on modified corn starch and grape juice. *Food Chemistry*, 292, 6–13.
- Zúñiga, M. M. G., Ding, R., Oh, E., Nguyen, T. B., Tran, T. T., Nam, J. D., & Suhr, J. (2024). Avocado seed starch utilized in eco-friendly, UV-blocking, and high-barrier polylactic acid (PLA) biocomposites for active food packaging applications. *International Journal of Biological Macromolecules*, 265(130), 837.